

Dake Bu

Ph.D. Student | Transformer In-Context Learning & LLM Post-Training | Continuous/Discrete Diffusion Generation & Sampling | Reinforcement Learning | Machine Learning Theory

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PROFESSIONAL SUMMARY

- Ph.D. student in Computer Science at CityUHK with research experience at A*STAR CFAR and RIKEN AIP / the University of Tokyo; author of multiple ICML and NeurIPS papers.
- Focuses on the theory and algorithms of large-scale machine learning systems, spanning emergent in-context learning in Transformers, agentic AI, LLM and diffusion post-training/inference, continuous and discrete generative models, reinforcement learning, and sampling. Combines algorithm design, rigorous theory, open-source systems, and reproducible experiments to improve foundation-model training and inference.

EDUCATION

City University of Hong Kong	Ph.D. in Computer Science; GPA 3.92/4.0; Supervisors: Qingfu Zhang, Hau-San Wong	Oct. 2023-Present
Xi'an Jiaotong University	B.Sc. in Information and Computational Science; GPA 86.62/100; Supervisors: Hui Li, Jian Sun	Aug. 2019-Jun. 2023

RESEARCH EXPERIENCE & SELECTED WORK (8 first-author / 2 second-author papers; ICML / NeurIPS)

ORGANIZATION	ROLE	DATES	RESEARCH FOCUS / RESPONSIBILITIES
RIKEN AIP	Research Intern; Deep Learning Theory Team; PI: Taiji Suzuki	Dec. 2024-Oct. 2025	Transformer pretraining, emergent in-context learning, and LLM post-training; studied how ICL and task vectors emerge through gradient-based training; designed curriculum-based reasoning post-training algorithms; characterized RLVR-induced distributional bias against rare correct paths and reasoning diversity; and developed algorithms balancing diversity and exploration.

ROLE	PAPER / PROJECT	STATUS
First author	Provable Sample Efficiency of Curriculum Post-Training for Transformer Reasoning [LLM POST-TRAINING & INFERENCE] Introduced two plug-and-play curricula, Depth-Increasing Curriculum and Hint-Decreasing Curriculum, with practical variants for RLVR fine-tuning and test-time scaling. Proved that both reduce the exploration complexity of rare correct trajectories from exponential to polynomial. On Qwen2.5-1.5B, consistently improved macro-average pass@K across three reasoning benchmarks from 26.9 to 29.0, Countdown from 36.0 to 42.6, and pass@1 from 5.2 to 9.4. [Paper] [GitHub]	ICML 2026
First author	Provably Transformers Harness Multi-Concept Word Semantics for Efficient In-Context Learning [TRANSFORMER PRETRAINING & EMERGENT ICL] Proved that a nonlinear Transformer with softmax attention and ReLU MLPs learns contextual concept selection from polysemous word representations, with exponentially decaying 0-1 error. Validated the predicted loss, attention/MLP projections, and OOD dynamics in 1,000 dimensions with a 10 ⁴ -word vocabulary and 5,000 test samples. [Paper] [GitHub]	NeurIPS 2024
First author	Provable In-Context Vector Arithmetic via Retrieving Task Concepts [TRANSFORMER PRETRAINING & EMERGENT ICL] Proved that nonlinear residual Transformers retrieve and compose latent task vectors to perform factual-recall vector arithmetic in context. Theory covers concept recombination and distribution shift; validated in 3,000 dimensions, 30 concept groups, and ICL / QA / QA-ICL training settings. [Paper] [GitHub]	ICML 2025
First author	Distributional Biases in Post-Training: A Markovian Analysis of Reasoning Trajectories [LLM POST-TRAINING & INFERENCE] Quantified "path squeezing" in RLVR: REINFORCE amplified common reasoning paths from 21.67% to 94.33% while suppressing rare correct paths from 8.07% to 3.43%. Designed an exploration-aware method that recovered rare-path mass to 36.10%, informing KL regularization and exploration design. [Paper] [GitHub]	ICML 2026 FoGen Spotlight
Second author	Multi-objective Linear Reinforcement Learning with Lexicographic Rewards [REINFORCEMENT LEARNING ALGORITHM DESIGN] Developed the first strict-priority multi-objective linear RL algorithm with regret guarantees: satisfy higher-priority hard constraints before optimizing lower-priority rewards. Achieved $\tilde{O}(\sqrt{K})$ regret for each objective and covered model misspecification and safety/cost-prioritized decision making. [Paper]	ICML 2025

ORGANIZATION	ROLE	DATES	RESEARCH FOCUS / RESPONSIBILITIES
CFAR, A*STAR	Research Intern; PI: Atsushi Nitanda	Dec. 2025-Present	Designed agentic harnesses and diffusion/sampling algorithms: built a Lean 4-based automated verification system; developed DLLM token-ordering modules validated on language, multimodal, and scientific sequence tasks; and proposed theoretically grounded training-free guidance algorithms for continuous diffusion models that outperform contemporary baselines in image generation.

ROLE	PAPER / PROJECT	STATUS
First author	ABEIS: State Preparation and Block Encoding for Quantum Computing [AGENTIC HARNESS SYSTEMS] Built a Lean 4-verifiable multi-agent platform for quantum-circuit state preparation and block encoding. A planner agent maintains candidate solutions and resource objectives, a coordinator decomposes proof DAGs, and worker agents generate circuits and invoke Lean verification. The system retains only compiling candidates, ranks them by gate count, depth, ancilla qubits, and oracle calls, and exports to Qiskit, QuantumKatas, and QASM, enabling an auditable candidate-evolution and deployment loop. [GitHub]	Open-source on GitHub; Paper in preparation
	DPRM: A Plug-in Doob h transform-induced Token-Ordering Module for Diffusion Language Models [DISCRETE DIFFUSION GENERATION & SAMPLING] Replaces only token ordering - without changing the backbone, training objective, or data pipeline - and selects the next token batch using confidence and process rewards. Integrated into nine language, multimodal, and scientific sequence-generation settings: GSM8K +16.8%, MATH Hard +8.1%, Countdown +12.8%, protein RMSD -17.0%, and DNA score 1.155 to 2.192. Demonstrates plug-and-play cross-task gains while preserving the original ordering as a switchable baseline. [Paper] [GitHub]	ICML 2026 FoGen Oral
Second author	Slowly Annealed Langevin Dynamics: Theory and Applications to Training-Free Guided Generation [CONTINUOUS DIFFUSION GENERATION & SAMPLING] Proposed SALD and Velocity-Aware SALD, reducing training-free guidance bias by slowing the motion of the target distribution. Derived a terminal-KL error decomposition and $O(\epsilon^{-6})$ discretization complexity to reach ϵ accuracy, and demonstrated stable improvements across image-generation tasks. [Paper] [GitHub]	Under review (arXiv preprint)
First author	Polynomial Convergence of Riemannian Mean-Field Langevin Dynamics: A Case Study on Single Index Model [SAMPLING & DYNAMICAL SYSTEMS] Reformulated training of a nonconvex single-index model as Riemannian mean-field Langevin dynamics on the sphere. Established explicit polynomial-time convergence bounds for multi-well, non-log-concave landscapes without global convexity, yielding verifiable conditions for sampling-based training of high-dimensional nonconvex models.	Under review

CITYUHK DOCTORAL RESEARCH (OUTSIDE INTERNSHIPS)

ROLE	PAPER / PROJECT	STATUS
First author	Provably Neural Active Learning Succeeds via Prioritizing Perplexing Samples [ACTIVE LEARNING & DATA SELECTION] Unified uncertainty and diversity sampling: both prioritize difficult samples containing yet-unlearned features. For the same generalization error, reduced labeling complexity by $\Theta(1/p)$ relative to random sampling; validated with Full, Random, Uncertainty, and Diversity strategies. [Paper] [GitHub]	ICML 2024

CORE EXPERTISE

- **Transformer Pretraining & Emergent In-Context Learning:** Study how Transformers trained with next-token prediction form reusable concept and task representations through gradient-based pretraining, enabling factual recall, contextual disambiguation, and task-vector composition; analyze how data, architecture, and optimization dynamics jointly produce ICL and compositional generalization.
- **LLM Post-Training & Inference Optimization:** Design curriculum-learning and theoretically grounded process-reward methods for sparse-reward post-training and RLVR-induced distribution shifts, improving sample efficiency while preserving rare correct reasoning paths and reasoning diversity.
- **Continuous & Discrete Diffusion Generation and Sampling:** Analyze convergence of Langevin/diffusion sampling and its use in training-free guidance for image and multimodal generation, token ordering, and parallel decoding. Experience spans language, vision, proteins, RNA, molecules, and DNA, with end-to-end work from theory to cross-modal validation.
- **Reinforcement Learning, Sampling & Dynamical Systems:** Develop strict-priority multi-objective RL and bandit algorithms, neural mean-field dynamics, non-log-concave sampling, and active data selection; combine algorithm design, convergence/sample-complexity proofs, and reproducible experiments for online decision-making and high-dimensional training dynamics.
- **Agentic Harness Systems:** Build end-to-end agentic pipelines with multi-agent decomposition, tool use, formal verification, failure feedback, and candidate selection, improving reliability, interpretability, and iteration efficiency without changing model architecture. Experienced with Lean-based automated theorem proving.

HONORS, SERVICE & SKILLS

- ICML 2026 Gold Reviewer Award; Area Chair: ICML 2026 FoGen, ICLR 2026 DeLTa; Reviewer: ICML / NeurIPS / ICLR (2024-2026).
- Scholarships: CityUHK Research Tuition Scholarship (Top 5%), Outstanding Academic Performance Award (Top 10%), Hong Kong Government PGS.
- Technical: Python, PyTorch, C/C++, Java, SQL, MATLAB, LaTeX; Languages: English (TOEFL 109), Mandarin (native), Japanese (basic).